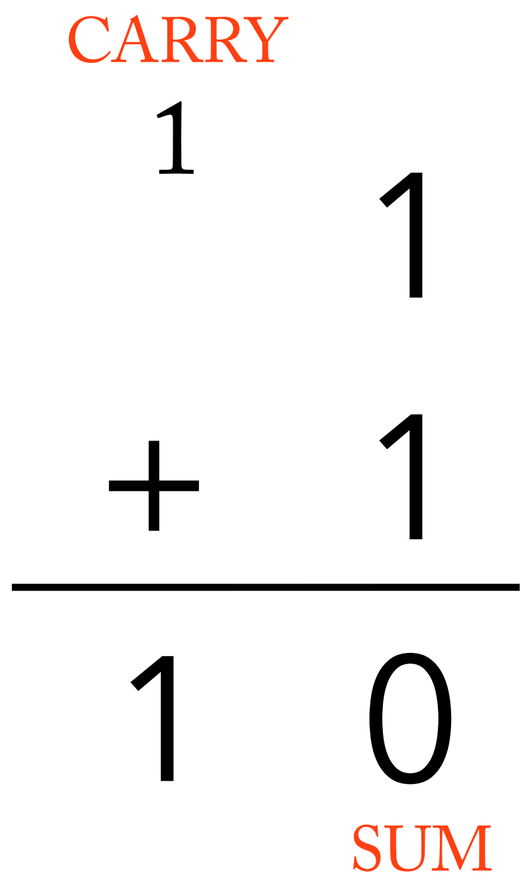
9. The Story of Binary Successor

The Problem: Binary Successor

For computer scientists, the computation of binary successor is the basis for nearly all more sophisticated computations.

Intro: Long Addition in the Simplest Case

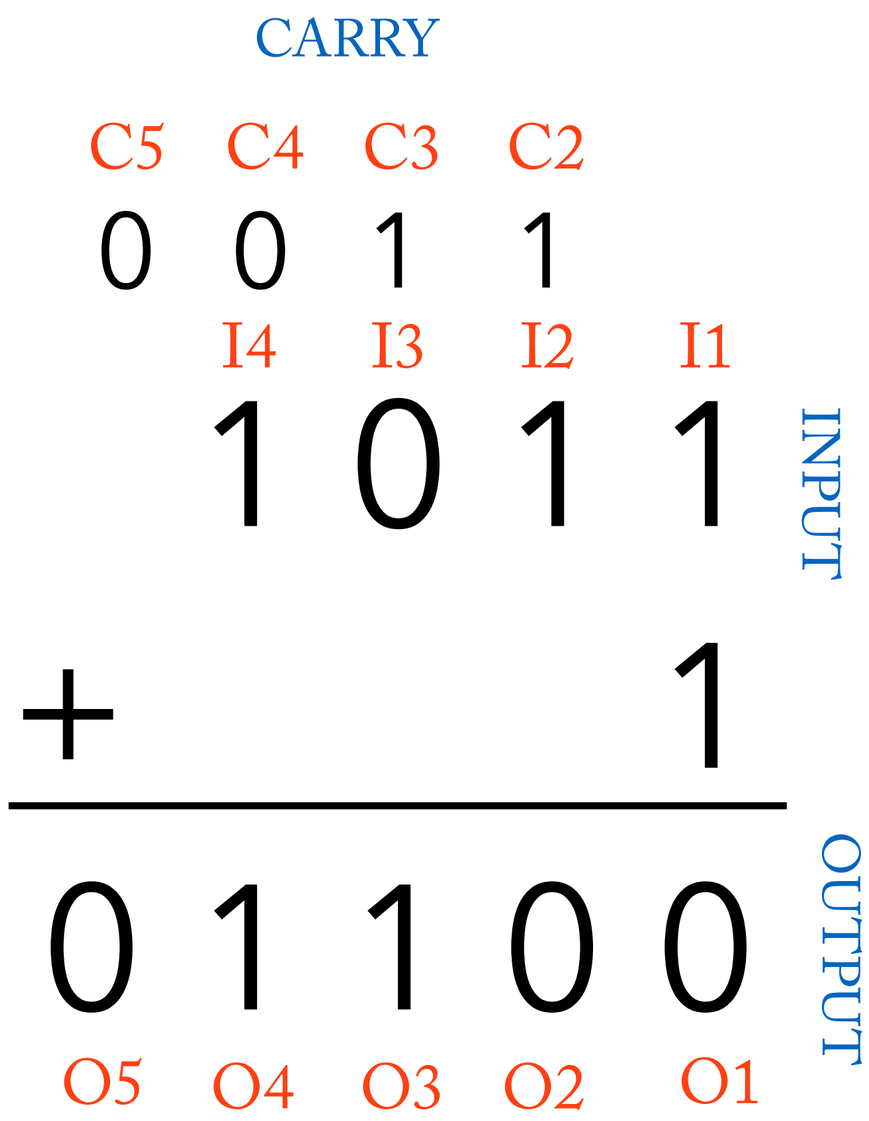


A *bit* = a single binary digit (or numeral).

The result of the initial sum is two bits long, which we break into parts.

The *first* bit is called the **SUM** bit (or S bit). The *second* bit is called the **CARRY** bit (or C bit).

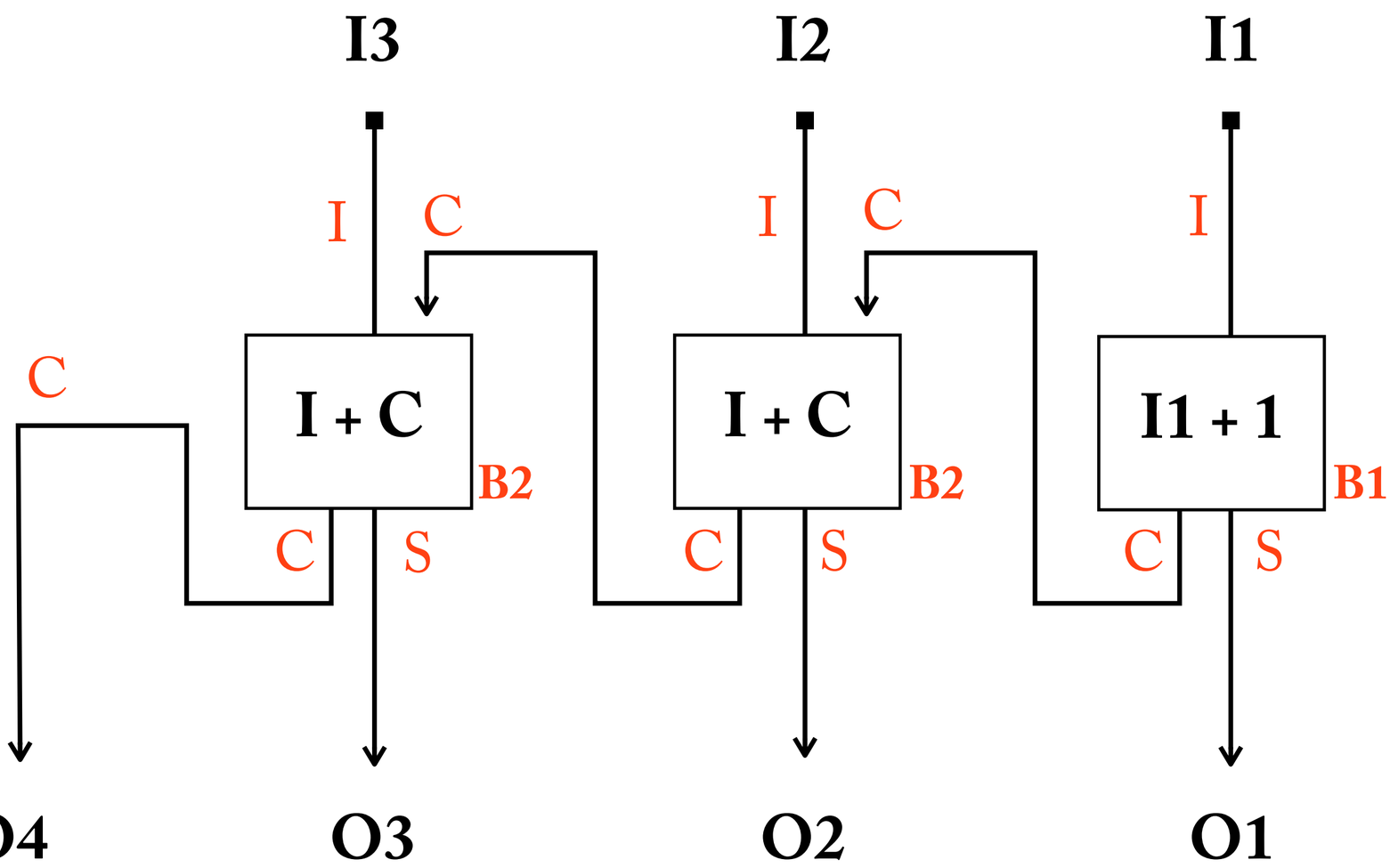
Step 1. Identify a method (carrying)



Binary successor algorithm

1. Compute I1 + 1.
   Results in a SUM bit and a CARRY bit.
2. O1 = SUM.
3. Compute I_{NEXT} + CARRY.
   Results in a new SUM bit and a new CARRY bit.
4. O_{NEXT} = SUM.
5. Repeat from 3, as long as there are inputs.
6. When there are no inputs left, O_{FINAL} = CARRY.

Step 2. Break it into parts



We could keep doing this forever...!

Step 3. Analyze the parts

Box 1: I1 +1

Input	Sum	Carry
1	0	1
0	1	0

Box 2: I +C

Input	Carry (IN)	Sum	Carry
1	1	0	1
1	0	1	0
0	1	1	0
0	0	0	0

Next you must design a circuit corresponding to each box….

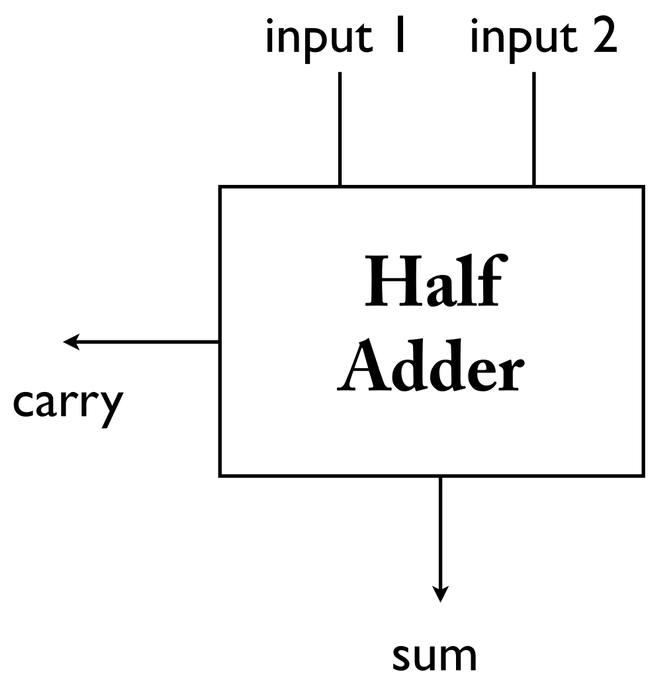
Interlude: Half-Adder

The Box 2 circuit is especially useful and important. It performs a key action: it takes any two bits as inputs, and outputs a SUM bit and a CARRY bit.

This mini-circuit is known as a "half adder". Half adders form the heart of any addition-based computation.

It is only *half* an adder, because it only adds two bits, rather than two bits *plus* a carry bit.

Schematically, it looks like this:



Always label your SUM and CARRY outputs!

Step 4: A schematic solution

